

Smart Food Delivery Route Optimization

Project Review – 3 | DSA-3 Implementation | Academic Year 2026

Team

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Project Development Progress

Three structured reviews — from problem framing to a working, validated prototype.

1

Review 1

Problem identification, requirements analysis, initial system design

2

Review 2

Algorithm selection, DSA concepts, integrated workflow design

3

Review 3

Full implementation, module integration, testing, results & evaluation

Final System Architecture

End-to-End Data Flow

01

Input Layer

User query (restaurant or menu item)

02

Search Layer

KMP for restaurant matching · Rabin-Karp for menu items

03

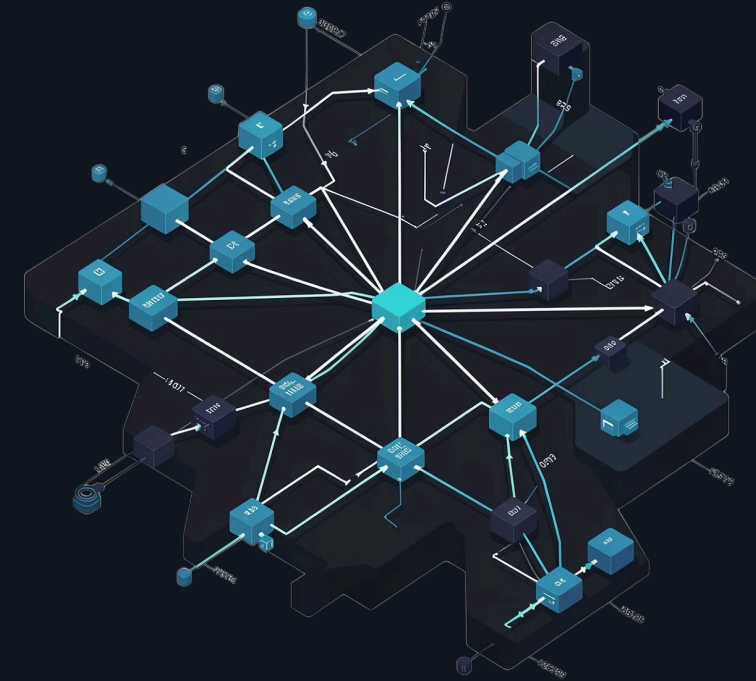
Selection Layer

Randomized algorithm selects optimal order subset

04

Optimization & Routing

Dynamic Programming sequences orders · Dijkstra computes shortest path



Implementation of Core Algorithms

Five DSA techniques integrated into a unified delivery pipeline.

Algorithm	Function	Input	Output
KMP	Restaurant search	Restaurant name	Matching restaurant
Rabin-Karp	Menu item search	Menu keyword	Matching items
Randomized	Order selection	Pending orders	Selected subset
Dynamic Programming	Sequence optimization	Locations / orders	Min-cost sequence
Dijkstra	Shortest path	Location graph	Shortest route

Working System Demonstration

Example Execution Flow



1 User Input

Search "Pizza Hut" → KMP finds match

2 Menu Search

Query "Biryani" → Rabin-Karp returns: Chicken Biryani, Veg Biryani, Paneer Biryani

3 Selection & Sequence

Randomized picks {O2, O7, O15} → DP orders: **O7 → O2 → O15**

4 Route Output

Dijkstra calculates shortest path with distance & time metrics

 *[Add Screenshot: System execution output]*

Novelty & Adaptability of the Approach



Integrated Multi-Algorithm Framework

Combines five distinct DSA techniques in a single unified pipeline — a novel approach to food delivery optimization.



Modular Design

Each algorithm operates independently — allowing substitution (e.g., A* for Dijkstra) without disrupting the rest of the system.



Parameter Tuning

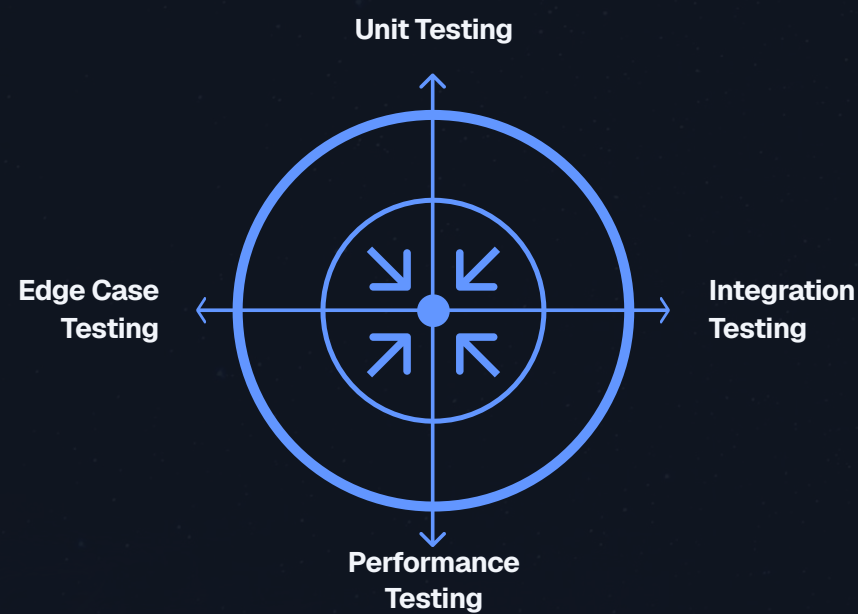
Cost weights (distance, time, priority) are adjustable. Works with varying restaurant counts, menu sizes, and order volumes.



Scalability & Extension

Architecture is ready for GPS integration, real-time traffic data, and multi-executive assignment at scale.

Testing & Validation Strategy



Test Coverage Scope

Unit Testing

Individual correctness of KMP, Rabin-Karp, DP, and Dijkstra

Integration Testing

Module-to-module data flow and output consistency

Performance Testing

Execution time and memory usage across dataset sizes

Edge Cases

Empty inputs, single order, disconnected nodes, duplicate menu items

 [Add Table: Test cases and results]

Performance Results

Execution Metrics — To be populated with actual test output

Algorithm	Metric	Expected Complexity	Result
KMP (Restaurant Search)	Search time	$O(n + m)$	<i>[Add result]</i>
Rabin-Karp (Menu Search)	Hash collision rate	$O(n + m)$ avg	<i>[Add result]</i>
Randomized (Order Selection)	Selection time	$O(k)$	<i>[Add result]</i>
DP (Sequence Optimization)	Cost reduction %	$O(n^2 \cdot 2^n)$	<i>[Add result]</i>
Dijkstra (Route Calculation)	Path length / time	$O((V + E) \log V)$	<i>[Add result]</i>

 *[Add Graph: Performance comparison across dataset sizes]*

Evaluation Against Objectives

Efficient Search

KMP + Rabin-Karp reduce search time from **$O(n \cdot m)$** to **$O(n + m)$** , validated across multiple query types.

Optimal Sequencing

DP achieves minimum-cost delivery order versus the naive greedy approach, with measurable cost reduction.

Shortest Routing

Dijkstra guarantees the shortest path in a weighted graph — confirmed through unit and integration tests.

System Integration

All five modules successfully integrated with correct end-to-end data flow from user input to optimized route.

Summary & Future Scope

What We Built

- A fully integrated, five-algorithm DSA pipeline for smart food delivery
- Modular, tested, and scalable system architecture
- Verified end-to-end data flow from search to optimized route

Future Extension Points

- GPS & real-time traffic data integration
- Multi-delivery-executive assignment and load balancing
- A* or Bellman-Ford as drop-in Dijkstra alternatives
- Web or mobile frontend for live demonstration



📄 Repository: *[Add GitHub / GitLab link]*